Random Numbers and Probability

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Random variable: A variable that can take multiple values, each with a certain probability

Let X be a random variable that can take values $x_1, x_2, \ldots, x_N$ with probabilities $Pr_1, Pr_2, \ldots, Pr_N$

This set of probabilities is called a probability distribution

By definition $\sum_{i=1}^{N} Pr_i = 1$

If X takes a continuous set of values between limits a and b, then we define a probability density $f(x)$.

$f(x)\,dx \equiv$ Probability that the value of X lies between x and $x+dx$

By definition $\int_a^b f(x)\,dx = 1$

Moments of distribution (M_n)

$$M_n = \sum_{i=1}^{N} x_i^n \, Pr(x_i)$$

1st moment $\equiv M_1 = \langle x \rangle \equiv$ average of X (mean) $= \sum_{i=1}^{N} x_i \, Pr\{x_i\}$

2nd moment $\equiv M_2 = \langle x^2 \rangle = \sum_{i=1}^{N} x_i^2 \, Pr\{x_i\}$

$$\begin{aligned}
\text{Variance} &= \sum_{i=1}^{N} (x_i - \langle x \rangle)^2 \, Pr\{x_i\} \\
&= \sum_{i=1}^{N} (x_i^2 - 2\langle x \rangle x_i + \langle x \rangle^2) \, Pr\{x_i\} \\
&= \sum_{i=1}^{N} x_i^2 \, Pr\{x_i\} - 2\langle x \rangle \sum_{i=1}^{N} x_i \, Pr\{x_i\} + \langle x \rangle^2 \sum_{i=1}^{N} Pr\{x_i\} \\
&= \langle x^2 \rangle - 2\langle x \rangle \langle x \rangle + \langle x \rangle^2
\end{aligned}$$

$\therefore$ Variance $\equiv \langle x^2 \rangle - \langle x \rangle^2$

For continuous random variables

$$M_1 = \langle x \rangle = \int_a^b x \, f(x)\,dx$$

$$M_2 = \langle x^2 \rangle = \int x^2 f(x)\,dx$$

Averages of functions of x : $g(x)$

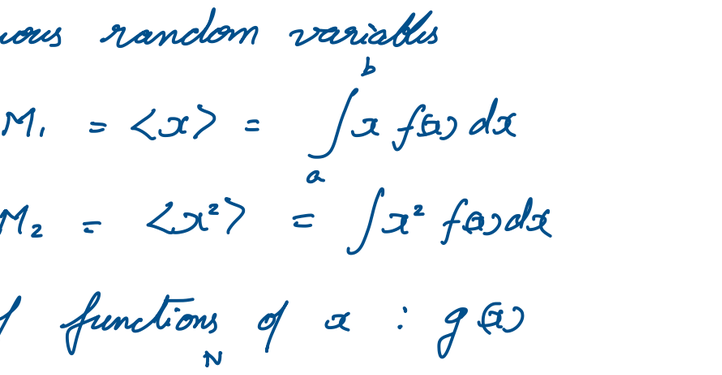
$$\langle g \rangle = \sum_{i=1}^{N} g(x_i) \, Pr\{x_i\}$$

$$\langle g \rangle = \int_a^b g(x) f(x)\,dx$$

Important probability density functions

1) Uniform density function

$f(x) = C \equiv$ constant in a given range (a, b) & zero outside this range



To obtain the value of C, we apply the normalization condition

$$\int_a^b f(x)\,dx = 1$$

$\therefore (b-a)\,C = 1$

$$\therefore \quad C = \frac{1}{b-a}$$

$$\therefore \quad f(x) = \frac{1}{b-a}$$

In a computer, we can generate sequences of random numbers that are distributed uniformly between 0 and 1. These are called "pseudo-random number generators" (PRNG)

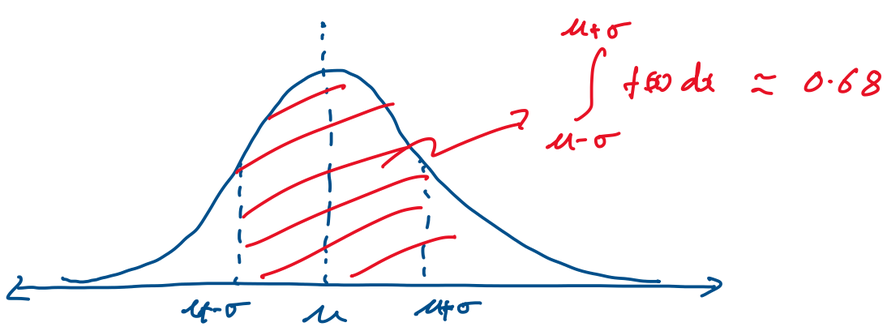
2) Normal distribution (Gaussian distribution)

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

μ = mean of the distribution
σ = standard deviation

$$\Rightarrow \quad \mu = \int_{-\infty}^{\infty} x \, \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 \, \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$



Computation of averages

Consider the function $g(x)$

$$\langle g \rangle = \sum_{i=1}^{N} g(x_i)\, P_x\{x_i\}$$

Two ways of computing the above average

1) Calculate $g(x_i)$ for every value of x_i

Compute the average from the formula, i.e.

$$\langle g \rangle = \sum_{i=1}^{N} g(x_i)\, P_x\{x_i\}$$

The drawback of this method is that it becomes inefficient when the probabity distribution is significant only for a few values of x_i and negligible for all other values.

e.g. Let $N = 10^6$ & $P_x\{x_i\} = \begin{cases} 0.1 & i = 100 \\ 0.8 & i = 101 \\ 0.1 & i = 102 \\ 0 & \forall \text{ all other } i \end{cases}$

It is clear that only three values of x_i contribute to $\langle g \rangle$

But this method spends a lot of effort in computing $g(x_i)$ which do not contribute to $\langle g \rangle$.

2) Generate a sequence of "M" random numbers according to the probability distribution $P_x\{x_i\}$

$$\therefore \langle g \rangle \approx \frac{1}{M} \sum_{j=1}^{M} g(x_j)$$

$$= \lim_{M \to \infty} \frac{1}{M} \sum_{j=1}^{M} g(x_j)$$

The above method samples only those values of x_i where the probability is significant.

This sampling is called "Importance Sampling".

Continuous variables

1) $$\langle g \rangle = \int_a^b g(x) f(x)\, dx$$

Compute the above integral mid-point rule

$$\langle g \rangle = \sum_{i=1}^{M} g(x_i) f(x_i) \Delta x \qquad \Delta x = \frac{b-a}{M}$$

x_i ↑ mid-point of our bin

2) Generate a sequence of M random number distributed according to $f(x)$.

$$\therefore \; g(x) \approx \frac{1}{M} \sum_{j=1}^{M} g(x_j)$$

To implement the 2nd method, we need an algorithm for generating a sequence of non-uniform random numbers.

We already know how to generate uniform random numbers using PRNG.

One can use this sequence to generate any sequence of non-uniform random numbers.

1) Invasion method
2) Acceptance – Rejection method

} Any textbook on probability will discuss this

In case of molecular simulations we use "Metropolis Algorithm" for generating non-uniform random numbers